Finitely-open inverse germs have free neighborhoods

Candidate full answer to the question after Proposition 5.5 of arXiv:1407.7551

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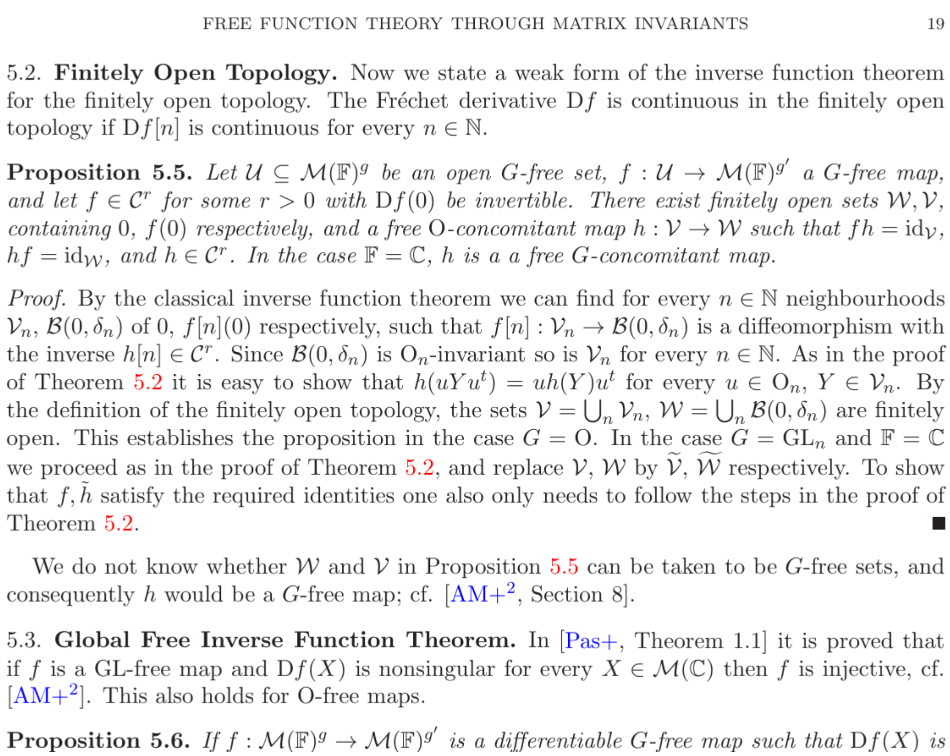
Status: candidate full affirmative answer. Under exactly the hypotheses of Proposition 5.5 of the source, its inverse neighborhoods can be chosen finitely open and G-free. The inverse is then a G-free C^r map. The proof needs no uniform radius: it constructs thin compatible neighborhoods inductively in the matrix size.

1 Question and answer

Let G be one of the groups considered in the source: $G = \mathrm{O}$ over $\mathbb{R}$, or $G = \mathrm{GL}$ over $\mathbb{R}$ or $\mathbb{C}$. A set in $\mathcal{M}(\mathbb{F})^g = \bigsqcup_{n \geq 1} M_n(\mathbb{F})^g$ is *finitely open* if its intersection with every matrix level is Euclidean open. A G-free set is closed under direct sums and simultaneous G-conjugation.

Klep and Špenko [1], Proposition 5.5, apply the classical inverse function theorem separately at every level to a G-free C^r map f with $Df(0)$ invertible. They obtain finitely open inverse neighborhoods, but ask whether those neighborhoods may be chosen G-free.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5.2. **Finitely Open Topology.** Now we state a weak form of the inverse function theorem for the finitely open topology. The Fréchet derivative $\mathrm{D}f$ is continuous in the finitely open topology if $\mathrm{D}f[n]$ is continuous for every $n \in \mathbb{N}$.

Proposition 5.5. *Let $\mathcal{U} \subseteq \mathcal{M}(\mathbb{F})^g$ be an open G-free set, $f : \mathcal{U} \to \mathcal{M}(\mathbb{F})^{g'}$ a G-free map, and let $f \in \mathcal{C}^r$ for some $r > 0$ with $\mathrm{D}f(0)$ be invertible. There exist finitely open sets $\mathcal{W}, \mathcal{V}$, containing 0, $f(0)$ respectively, and a free O-concomitant map $h : \mathcal{V} \to \mathcal{W}$ such that $fh = \mathrm{id}_{\mathcal{V}}$, $hf = \mathrm{id}_{\mathcal{W}}$, and $h \in \mathcal{C}^r$. In the case $\mathbb{F} = \mathbb{C}$, h is a a free G-concomitant map.*

Proof. By the classical inverse function theorem we can find for every $n \in \mathbb{N}$ neighbourhoods $\mathcal{V}_n$, $\mathcal{B}(0, \delta_n)$ of 0, $f[n](0)$ respectively, such that $f[n] : \mathcal{V}_n \to \mathcal{B}(0, \delta_n)$ is a diffeomorphism with the inverse $h[n] \in \mathcal{C}^r$. Since $\mathcal{B}(0, \delta_n)$ is O_n-invariant so is $\mathcal{V}_n$ for every $n \in \mathbb{N}$. As in the proof of Theorem 5.2 it is easy to show that $h(uYu^t) = uh(Y)u^t$ for every $u \in \mathrm{O}_n$, $Y \in \mathcal{V}_n$. By the definition of the finitely open topology, the sets $\mathcal{V} = \bigcup_n \mathcal{V}_n$, $\mathcal{W} = \bigcup_n \mathcal{B}(0, \delta_n)$ are finitely open. This establishes the proposition in the case $G = \mathrm{O}$. In the case $G = \mathrm{GL}_n$ and $\mathbb{F} = \mathbb{C}$ we proceed as in the proof of Theorem 5.2, and replace $\mathcal{V}$, $\mathcal{W}$ by $\widetilde{\mathcal{V}}$, $\widetilde{\mathcal{W}}$ respectively. To show that $f, \tilde{h}$ satisfy the required identities one also only needs to follow the steps in the proof of Theorem 5.2. ■

We do not know whether $\mathcal{W}$ and $\mathcal{V}$ in Proposition 5.5 can be taken to be G-free sets, and consequently h would be a G-free map; cf. [AM+[2], Section 8].

5.3. **Global Free Inverse Function Theorem.** In [Pas+, Theorem 1.1] it is proved that if f is a GL-free map and $\mathrm{D}f(X)$ is nonsingular for every $X \in \mathcal{M}(\mathbb{C})$ then f is injective, cf. [AM+[2]]. This also holds for O-free maps.

Proposition 5.6. *If $f : \mathcal{M}(\mathbb{F})^g \to \mathcal{M}(\mathbb{F})^{g'}$ is a differentiable G-free map such that $\mathrm{D}f(X)$ is*

Figure 1: Source evidence: page 19, Section 5.2 of arXiv:1407.7551. The source states: "We do not know whether $\mathcal{W}$ and $\mathcal{V}$ in Proposition 5.5 can be taken to be G-free sets"; it notes that this would make the inverse G-free.

2 Proof intuition

The obstacle is not the inverse function theorem at a single matrix level; it is choosing all levelwise neighborhoods compatibly with direct sums. The key observation is that this compatibility is only pairwise. At a finite direct sum, block sign conjugations split the derivative into the diagonal channels and one off-diagonal channel for each pair of summands. Thus invertibility at all two-summand combinations implies invertibility at every finite direct sum.

This permits an induction on matrix size. At level n, all direct sums forced by the already chosen smaller neighborhoods form a compact set after orthogonal/unitary conjugation. Pairwise invertibility is open, so a thin invariant tube around that compact set remains regular and compatible with all earlier levels. These tubes form a finitely open compact-group-free domain. Taking its full similarity envelope supplies the GL cases. Nonsingularity on the resulting free domain then yields injectivity by the standard two-point block identity, and the ordinary levelwise inverse theorem finishes the proof.

Theorem 1 (Free finitely-open inverse theorem)**.** *Let $\mathcal{U} \subseteq \mathcal{M}(\mathbb{F})^g$ be a finitely open G-free set containing zero, and let*

$$f : \mathcal{U} \longrightarrow \mathcal{M}(\mathbb{F})^g$$

be a G-free C^r map, $r \geq 1$. Suppose $Df(0_n)$ is invertible at every level n. Then there are finitely open G-free sets $\mathcal{W} \subseteq \mathcal{U}$ and $\mathcal{V} = f(\mathcal{W})$, containing 0 and $f(0)$ respectively, such that

$$f|_{\mathcal{W}} : \mathcal{W} \longrightarrow \mathcal{V}$$

is a bijection. Its inverse $h : \mathcal{V} \to \mathcal{W}$ is a G-free C^r map. Consequently the question after Proposition 5.5 has an affirmative answer.

No uniform lower bound on the levelwise inverse radii is asserted or needed. The construction uses small tubes around all direct sums already forced at a given level.

3 Pairwise compatibility controls every direct sum

Write $X \sim Y$ if $Df(X \oplus Y)$ is invertible. This is an open relation at each fixed pair of levels because Df is levelwise continuous. We also call a point X *regular* when $Df(X)$ is invertible.

Lemma 2 (Pairwise block lemma)**.** *Let $X^1, \dots, X^s$ be points, possibly of different matrix sizes. If every X^i is regular and $X^i \sim X^j$ for $i \neq j$, then*

$$Df(X^1 \oplus \cdots \oplus X^s)$$

is invertible.

Proof. Put $A = X^1 \oplus \cdots \oplus X^s$. For $\varepsilon = (\varepsilon_1, \dots, \varepsilon_s) \in \{\pm 1\}^s$, the block sign matrix

$$S_\varepsilon = \operatorname{diag}(\varepsilon_1 I, \dots, \varepsilon_s I)$$

belongs to every group under consideration and fixes A. Differentiating the equivariance identity at A shows that $Df(A)$ commutes with conjugation by every S_ε. Hence the tangent space splits into invariant character spaces: the block-diagonal space and, for every unordered pair $i < j$, the space supported in the two off-diagonal blocks (i,j) and (j,i).

On the diagonal space, direct-sum preservation identifies the restrictions with $Df(X^i)$. On the (i,j) character space, the restriction is exactly the corresponding off-diagonal restriction of $Df(X^i \oplus X^j)$. The latter operator also respects the diagonal/off-diagonal splitting. Since $Df(X^i)$, $Df(X^j)$, and $Df(X^i \oplus X^j)$ are invertible, that off-diagonal restriction is invertible. Thus every invariant summand of $Df(A)$ is invertible, proving the claim. □

The lemma remains true with repeated summands. In particular, compatibility with 0_1 implies compatibility with $0_k = 0_1^{\oplus k}$ for every k.

4 Compact-group induction

First work with the compact subgroup

$$\mathsf{K}_n = \begin{cases} \mathrm{O}_n, & \mathbb{F} = \mathbb{R}, \\ \mathrm{U}_n, & \mathbb{F} = \mathbb{C}. \end{cases}$$

For $G = \mathsf{O}$ this is the original group; for $G = \mathrm{GL}$ it is a subgroup. We construct precompact, K_n-invariant open sets $B_n \subseteq \mathcal{U}[n]$ with the following properties:

1. $0_n \in B_n$ and every point of $\overline{B_n}$ is regular;
2. if $i, j \leq n$, then every $X \in \overline{B_i}$ and $Y \in \overline{B_j}$ satisfy $X \sim Y$;
3. if $i + j = n$, then $B_i \oplus B_j \subseteq B_n$.

Closures are taken in their finite-dimensional levels.

Suppose $B_1, \ldots, B_{n-1}$ have been chosen. Let C_n consist of 0_n and all K_n-conjugates of all direct sums

$$X^1 \oplus \cdots \oplus X^s, \qquad X^a \in \overline{B_{n_a}}, \quad n_a < n, \quad \sum_a n_a = n.$$

There are only finitely many ordered partitions of n, and the groups and the earlier closures are compact; therefore C_n is compact. It lies in $\mathcal{U}[n]$, because $\mathcal{U}$ is free. By the induction hypotheses and Lemma 2, all its points are regular, every two points of C_n are compatible, and every point of C_n is compatible with every earlier $\overline{B_i}$. Here 0_n causes no exception: decompose it into n copies of 0_1.

Regularity and compatibility are open conditions. Equip each level with a K-invariant Euclidean norm and set

$$N_n(\delta) = \{X : \mathrm{dist}(X, C_n) < \delta\}.$$

These tubes are automatically K_n-invariant. Compactness of C_n, of $C_n \times C_n$, and of $C_n \times \overline{B_i}$ for the finitely many $i < n$ gives $\delta > 0$ so small that $\overline{N_n(\delta)} \subset \mathcal{U}[n]$, all points of that closure are regular, every two of them are compatible, and every one of them is compatible with every earlier $\overline{B_i}$. (Otherwise a sequence with $\delta \downarrow 0$ would converge to a point of one of those compact sets outside the corresponding open condition.) Set $B_n = N_n(\delta)$. Since C_n contains every earlier direct sum of total size n, property (3) also holds. This completes the induction.

Now put $\Omega = \bigsqcup_{n \geq 1} B_n$. It is finitely open, K-invariant, contains zero at every level, and is closed under direct sums. Every point of Ω is regular.

If $G = \mathrm{O}$, set $\mathcal{W} = \Omega$. If $G = \mathrm{GL}$, take the full similarity envelope

$$\mathcal{W}[n] = \{SXS^{-1} : X \in B_n,\ S \in \mathrm{GL}_n(\mathbb{F})\}.$$

It is open (a union of linear images of B_n), closed under direct sums, and contained in $\mathcal{U}$. Thus it is a finitely open G-free set. Differentiating equivariance gives

$$Df(SXS^{-1})[SHS^{-1}] = S\,Df(X)[H]S^{-1},$$

so regularity passes to the similarity envelope. In all cases we have constructed a finitely open G-free neighborhood $\mathcal{W} \subseteq \mathcal{U}$ on which Df is everywhere invertible.

5 Injectivity, openness, and the inverse

For completeness, nonsingularity on a free domain forces injectivity by the standard two-point block argument. If $f(X) = f(Y)$ at one level, equivariance and direct-sum preservation give

$$Df(X \oplus Y) \begin{bmatrix} 0 & X - Y \\ X - Y & 0 \end{bmatrix} = 0.$$

The displayed matrix is understood componentwise in the g variables. Since $Df(X \oplus Y)$ is invertible, $X = Y$. Points at different levels cannot have the same graded value. Hence $f|_{\mathcal{W}}$ is injective.

At every level, Df is invertible throughout $\mathcal{W}[n]$. The classical inverse function theorem says that $f(\mathcal{W}[n])$ is open and that the inverse is C^r locally; injectivity glues these local inverses into a global levelwise C^r inverse. Direct sums and G-conjugations are inherited from f by uniqueness, so $\mathcal{V} = f(\mathcal{W})$ is G-free and h is a G-free map. This proves Theorem 1.

6 Literature and scope

The uniformly-open inverse theorem in the source and the operator-space nc inverse theorem of Abduvalieva–Kaliuzhnyi-Verbovetskyi [2] use a uniform nc radius (and, in the latter setting, completely bounded inverse control). Pascoe's fine theorem, as presented by Agler–McCarthy [3], gives an inverse on a fine nc domain when the derivative is nonsingular *everywhere* on that domain. The compact induction above supplies the missing domain from nonsingularity only at zero and uses only levelwise C^1 continuity; it therefore also covers the nonanalytic orthogonal case.

Exact-question, citation, nc-germ, and fine-topology searches through 12 August 2026 found no later paper explicitly resolving the source's question. The argument should nevertheless receive specialist review, especially its compact-neighborhood induction, before being claimed as new.

References

[1] I. Klep and Š. Špenko, *Free function theory through matrix invariants*, Canad. J. Math. 69 (2017), 408–433; arXiv:1407.7551.

[2] G. Abduvalieva and D. S. Kaliuzhnyi-Verbovetskyi, *Implicit/inverse function theorems for free noncommutative functions*, J. Funct. Anal. 269 (2015), 2813–2844; arXiv:1502.05254.

[3] J. Agler and J. E. McCarthy, *The implicit function theorem and free algebraic sets*, Trans. Amer. Math. Soc. 368 (2016), 3157–3175.